Rotaract Club of Xiamen University Malaysia | Rotary Club of Kuala Lumpur Dynamics | Rotary Club of Tun Razak Exchange

SDC Hackathon

Phase 1 : Preliminary Project Proposal

Hosted by Xiamen University Malaysia (XMUM) AI Club

Submission Deadline: 25 October 2026

Instructions: Complete every field below and submit as a single PDF document named "SDC2026_TeamName_Preliminary.pdf" before the deadline above. Answers must comply with the word limits stated for each open-ended question. Incomplete submissions may not be eligible for review.

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1. Team & Project Overview

Team Name	X- Force
Selected Domain	DOMAIN 2
Selected SDG Track (1–17)	SDG 3, SDG 12, SDG 13
Project / Solution Name	SURPLUSLink AI
Name (Leader)	PREESHA DIVINE SIMON
Email & Phone (Leader)	J26046777@newinti.student.edu.my , 0189000665
University Affiliation(s)	INTI INTERNATIONAL COLLEGE SUBANG
Number of Team Members	4

2. Executive Summary & Problem Statement

2.1 The Problem Context

Describe the specific, real-world challenge your team is tackling. Who is affected by this problem, and why does it matter? Highlight the critical social, environmental, or economic issues involved.

[Maximum 250 words]

Our team is tackling the problem of food waste and financial loss among small food businesses in Malaysia. Cafés, bakeries, and caterers sometimes have extra ingredients because of cancelled events or over-ordering. These ingredients may be thrown away when they are close to expiring, causing businesses to lose money.

At the same time, other nearby businesses may need those ingredients but still have to pay higher prices because they do not know where to find extra stock.

This problem affects small business owners financially and contributes to food waste, which can harm the environment. It also makes it harder for small businesses to manage their costs.

To solve this problem, our team proposes SurplusLink AI, a platform that connects businesses with extra ingredients to nearby businesses that need them. This helps sellers recover some of their costs, allows buyers to save money, and reduces unnecessary food waste.

2.2 The Proposed AI Solution

Provide a high-level overview of your proposed application. How does it apply artificial intelligence to solve the problem stated above? Briefly mention the tools or technical skills gained from the SDC Phase 1 workshops (e.g., GenAI APIs, database architecture, cloud hosting) that you intend to leverage.

[Maximum 300 words]

SurplusLink AI is an AI-powered platform designed to reduce food waste and financial losses among small food businesses in Malaysia. It connects businesses with extra ingredients, such as cafés, bakeries, and caterers, to nearby businesses that need them before they expire.

The platform uses artificial intelligence (AI) to make the process easier and faster. AI-assisted listing extraction helps identify ingredient names, quantities, and expiry dates from sellers' listings. The Culinary Matchmaker recommends suitable buyers based on their ingredient needs and location. Dynamic pricing suggests affordable prices based on the ingredients'

original cost and remaining shelf life. A Safety Verifier also helps check important information, such as expiry dates and storage requirements, before an exchange is approved.

To develop the prototype, our team intends to apply technical skills gained from the SDC Phase 1 workshops, including using Generative AI APIs to support AI features, database architecture to store business listings and ingredient information, and cloud hosting to make the platform accessible online. We also plan to use a simple, user-friendly interface so business owners can list and find surplus ingredients easily.

By connecting sellers and buyers, SurplusLink AI aims to help sellers recover some of their costs, allow buyers to obtain ingredients at lower prices, and reduce unnecessary food waste. Our Phase 2 goal is to build and test a working prototype, gather user feedback, and improve its safety, accuracy, and usability.

3. Technical Architecture & Implementation Plan

3.1 Conceptual Tech Stack

List the technologies, frameworks, APIs, and databases you plan to use to build a functional, deployed prototype. Describe (or attach as an appendix) a simple system block diagram showing how the workshop elements connect into one architecture.

Did your project utilize any official SDC sponsor-provided GenAI API or Cloud Platform? If yes, specify which one and how it is used.

Technologies, frameworks, APIs, and databases:

Frontend: React 18, Vite 5, JavaScript (JSX), plain CSS

Backend: Node.js with Express (lightweight proxy server)

Authentication: Firebase Authentication (email/password)

Database: Google Cloud Firestore (asia-southeast1/Singapore region)

AI: Google Gemini API (model gemini-3.8-flash), accessed server-side only through the Express proxy

Hosting/Deployment: Replit

System block diagram: See submission/architecture.png in the project repository (appendix). It shows: Browser (React frontend) → Express server on Replit → splits to (a) Google Gemini API for AI features and (b) Firebase Authentication/Cloud Firestore for user accounts and data storage.

No. The project used the team's own Google Firebase project (Authentication and Cloud Firestore) and the team's own Google Gemini API key. No official SDC sponsor-provided GenAI API or cloud platform credential was used.

3.2 Design-Thinking & Deployment Strategy

Explain how your team applied design-thinking methodology to ensure this solution is technically viable, user-friendly, and ready for real deployment by Phase 2.

[Maximum 200 words]

Our team applied design thinking to develop SurplusLink AI by focusing on the needs of small food businesses in Malaysia.

Empathise: We identified the challenges faced by cafés, bakeries, and caterers, such as losing money from unused ingredients and struggling to find affordable supplies.

Define: We focused on the problem of food waste and financial loss caused by surplus ingredients.

Ideate: We proposed SurplusLink AI, a platform that connects businesses with extra ingredients to nearby businesses that need them.

Prototype: We planned key features, including AI-assisted listings, ingredient matching, dynamic pricing, and safety checks to demonstrate how the platform could work.

Test: The prototype can be tested with small food businesses to gather feedback and improve its usability, accuracy, and safety.

By Phase 2, our goal is to develop a working prototype, improve it based on user feedback, and check that its features are practical and safe. This will help prepare SurplusLink AI for real-world deployment.

4. Preliminary Environmental & SDG Impact Framework

To align with SDC's focus on meaningful global sustainability, establish your project's impact baseline before building the prototype. Use estimated data, published research, or public benchmarks where exact figures are unavailable. Provide the citation of any sources used.

The target, baseline, and KPI you provide here will be used by judges during shortlisting to assess ***Innovation and Creativity*** *as well as* ***Feasibility of build****. Vague or unsupported projections may weaken your case for advancing to Phase 2.*

Metric Category	**Your Response**
Target SDG Sub-Target(s) *Identify the exact UN SDG sub-target your project addresses* *e.g., Target 11.6 "Reduce the adverse per-capita environmental impact of cities, focusing on air quality and waste management."*	SDG 12.3: By 2030, halve global food waste at retail and consumer levels and reduce food losses along production and supply chains.
Current Problem Baseline *What is the starting condition of the problem, expressed as a number?* *e.g., "The target infrastructure currently experiences 35% energy inefficiency due to unoptimized scheduling."*	Globally, approximately 19% of food available to consumers was wasted at retail, food service, and household levels in 2022. This shows the scale of the problem, although it is not specific to Malaysian small businesses.
Core Measurable KPI *The specific unit you will track* *e.g., % reduction in power consumption, processing-latency improvement, or kg of resource waste avoided.*	Kilograms (kg) of surplus ingredients rescued from being wasted through completed transactions on SurplusLink AI.
Phase 2 Impact Target *Your realistic, percentage-based target by the final prototype presentation* *e.g., "20% improvement in computational resource use."*	Aim to rescue at least 50 kg of surplus ingredients through completed exchanges during the prototype testing period. This is a proposed team target, not a proven result.

5. Eligibility & Readiness Checklist

Tick all boxes that apply before submitting.

- ☒ All team members have completed the mandatory GenAI, Database, and Cloud Deployment upskilling workshops.
- ☒ The project concept addresses a verified real-world problem mapped directly to one or more UN SDGs.
- ☒ The proposed solution targets a functional, deployed application rather than a theoretical business case study.
- ☒ All fields in this template have been completed within the word limits stated.

6. IP & Originality Declaration

6.1 Pre-Existing / Prior-Owned Work

Does this submission include any code, design, research, or other work that any team member created or owned before the start of this hackathon?

☐ Yes ☑ No

If yes, briefly describe what was pre-existing and which team member(s) owned it:

6.2 Third-Party Code, Datasets, or Models

Does your project use any third-party code libraries, open-source frameworks, public/private datasets, or pre-trained AI models that your team did not create?

☑ Yes ☐ No

If yes, list them below (open-source use is permitted provided it is declared and properly licensed/attributed):

Name	Type (code/dataset/model)	License
React and react DOM	**Code(library)**	**MIT License**
Vite +@vitejs/plugin-react	**Code (build tool)**	**MIT License**
Firebase JS SDK (Aunthentication + Firestore)	**Code(library)**	**Apache License 2.0**
Firebase Aunthentication and Cloud Firestore	**Cloud service (BaaS)**	**Google Firebase Terms of service**
Google Gemini API	**Pre-trained AI model (accessed via API)**	**Google APIs Terms of service / gemini API additional Terms**

Express.js	**Code (library, server proxy)**	**MIT License**
dotenv	**Code (library)**	**BSD-2-Clause License**

All third-party code is used via standard public npm packages and public APIs, properly declared in package.json. No proprietary or restricted third-party code was used. The project's own code is licensed under Apache License 2.0

6.3 Employment-Related IP Conflicts

Is any team member currently bound by an employment, internship, or consulting agreement under which their employer could claim ownership of work they create during this hackathon?

☐ Yes ☒ No

If yes, please describe the nature of the agreement and which member it concerns:

7. Declaration

By submitting this proposal, the team leader confirms that the information provided is accurate to the best of the team's knowledge and that the project will be original work created during the SDC 2026 hackathon period.

[Signature] 

[Date] 25/10/2026